

AI SIEM Platform

Basic Project Questions And Answers

Use this PDF for project explanation, viva preparation, and demo practice. The questions cover how the SOC platform works, how to run it, and how rules, alerts, incidents, and AI analysis are used.

1. Project Overview

No.	Question	Answer
1	What is the name of this project?	The project is an AI SIEM Platform, a Security Operations Center lab for monitoring logs, alerts, incidents, and AI-based analysis.
2	What is the main purpose of this project?	Its purpose is to collect security logs, detect suspicious activity using rules, generate alerts, manage incidents, and support SOC analysis.
3	What is SIEM?	SIEM means Security Information and Event Management. It collects and analyzes security events from systems, networks, and applications.
4	What is a SOC?	SOC means Security Operations Center. It is the team or platform responsible for monitoring, investigating, and responding to security threats.
5	Why is this project useful?	It demonstrates a real-world security monitoring workflow, including log collection, alert generation, incident handling, and analysis.

2. Architecture And Technologies

No.	Question	Answer
1	Which technologies are used in the frontend?	The frontend uses React, Vite, Tailwind CSS, Recharts, Axios, Zustand, and Socket.IO client.
2	Which technologies are used in the backend?	The backend uses Node.js, Express.js, Socket.IO, JWT authentication, bcrypt, PostgreSQL, Redis, and rule/AI services.
3	Which database is used?	PostgreSQL is used to store users, logs, alerts, incidents, rules, and analysis history.
4	Why is Redis used?	Redis is used for supporting background processing, queues, rate limiting, and fast temporary data operations.
5	Why is Docker used?	Docker is used to run frontend, backend, PostgreSQL, and Redis as separate containers with one Docker Compose setup.
6	What are the main containers in this project?	The main containers are frontend, backend, postgres, and redis.

3. Running The Project

No.	Question	Answer
1	How do you start the project?	Run docker compose up -d from the project root after setting the .env file and running database migrations/seeding.
2	What is the frontend URL?	The frontend runs at http://localhost:5173.
3	What is the backend URL?	The backend runs at http://localhost:3001.
4	How do you check backend health?	Open http://localhost:3001/health in the browser or call it using PowerShell/cURL.
5	What are the default login credentials?	Email: admin@siem.local and Password: changeme123.
6	How do you generate demo attack data?	Run .\tools\simulate-attacks.ps1 from PowerShell inside the project folder.

4. Logs, Rules, And Alerts

No.	Question	Answer
1	What is a log in this project?	A log is a security event such as failed login, SQL injection attempt, firewall event, endpoint alert, or suspicious process activity.
2	How are logs ingested?	Logs can be ingested through REST API endpoints, demo scripts, or syslog listeners.
3	What is a rule?	A rule is detection logic that checks logs for suspicious patterns or conditions.
4	Where can rules be created?	Rules can be created from the Rules page in the frontend.
5	What are common rule types in this project?	Common rule types include pattern, regex, threshold, field_match, and correlation.
6	What is an alert?	An alert is generated when a log matches an enabled detection rule.
7	What is the alert status flow?	The usual flow is open, acknowledged, and resolved.
8	What does acknowledge mean?	Acknowledge means the SOC analyst has seen the alert and started investigation.
9	What does resolve mean?	Resolve means the alert investigation is completed and the alert is closed.
10	What is bulk acknowledge?	Bulk acknowledge lets the analyst select many alerts and acknowledge them together.

5. Incidents And Investigation

No.	Question	Answer
1	What is an incident?	An incident is a larger investigation case created from one or more related alerts or suspicious events.
2	What is the difference between an alert and an incident?	An alert is a single detection. An incident is a case that groups related alerts and response activity.

No.	Question	Answer
3	Where can incidents be created?	Incidents can be created from the Incidents page.
4	What details are needed to create an incident?	The user enters a title, description, and severity.
5	Give one example incident title.	Example: Simulated Attack Campaign - Brute Force and Credential Dumping.
6	Why are incidents important?	They help track investigation progress, response actions, severity, and closure of security cases.

6. AI Analysis

No.	Question	Answer
1	What is the AI Analysis page used for?	It is used to analyze logs, alerts, and investigation context for triage, threat hunting, forensics, anomaly review, and IOC extraction.
2	What are the available AI analysis types?	The available types include AI Runbook, Threat Hunt, Anomaly, Forensics, Triage, and IOC Extraction.
3	What should be pasted into the AI Analysis text box?	The analyst can paste alert messages, log details, incident notes, source IPs, usernames, hostnames, or suspicious activity context.
4	What is IOC extraction?	IOC extraction means identifying indicators of compromise such as IP addresses, usernames, hostnames, file names, hashes, and attack techniques.
5	Does the project require an AI API key to work?	No. If an Anthropic API key is not set, the project can still use an offline SOC-style analysis mode.

7. Security And Demo Questions

No.	Question	Answer
1	How is authentication handled?	Authentication uses login credentials and JWT tokens to protect API routes.
2	What roles are supported?	The project supports roles such as viewer, analyst, and admin.
3	What demo attacks are shown?	The demo includes brute force login, SQL injection, suspicious login success after failures, and credential dumping.
4	What is brute force detection?	It detects repeated failed login attempts from the same user or source.
5	What is SQL injection detection?	It detects suspicious SQL patterns such as UNION SELECT or 1=1 in web request logs.
6	What is credential dumping detection?	It detects tools or behavior related to stealing credentials, such as Mimikatz or LSASS dumping.
7	How can you explain the full workflow in a demo?	Logs are collected, rules check the logs, alerts are created, analysts acknowledge alerts, serious cases become incidents, and AI analysis helps investigation.
8	What is one future improvement for this project?	Future improvements can include HTTPS, stronger password rotation, more automated tests, backup scheduling, and production monitoring.

Quick Demo Explanation

Short answer for presentation: This project is an AI-powered SIEM/SOC platform. It collects logs, applies detection rules, generates alerts, lets analysts manage incidents, and uses AI analysis to support triage, threat hunting, forensics, and IOC extraction.